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Fiscal Deficits with Heterogeneous Agents

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All codes for results and figures in this thesis are available on my Github repository <https://github.com/dunst04/Bachelor>. I want to thank dr Piotr Żoch for materials from his Quantitative Economics course and my advisor dr Paweł Kopiec for help and patience during writing of this thesis.

1. Introduction

Public debt-to-GDP ratios across advanced economies have climbed to levels not seen outside wartime, while safe real interest rates spent much of the period below the growth rate of output. This configuration, often summarised as $r < g$, sits uneasily with the textbook logic in which debt must ultimately be repaid out of future surpluses, and it has reopened a basic question: how large a permanent deficit can a government sustain, and what determines that limit? This thesis studies that question through the lens of household heterogeneity. The central idea is that when households face uninsurable income risk and cannot borrow freely, they accumulate precautionary savings, and government debt supplies the safe asset they demand. The size of the deficit a government can run forever, which is not possible in representative agent framework, is therefore pinned down in general equilibrium by how strongly households want to save. My aim is to make this mechanism explicit in a transparent incomplete-markets model, to quantify the maximum permanent deficit it permits, and to ask how that fiscal space depends on the tax system and how the resulting equilibria compare in welfare terms.

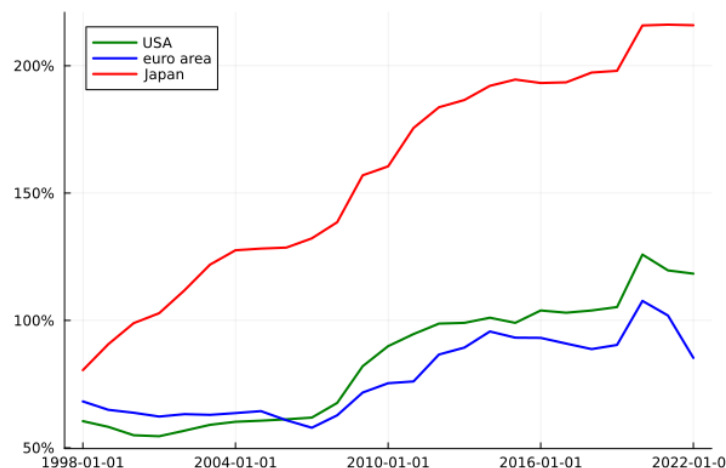


Figure 1: Debt to GDP ratio in USA, Japan and euro area, data source: FRED.

In recent years Heterogeneous Agent New Keynesian models (HANK in short) became a household name in modelling monetary and fiscal policy and business cycle. They are a natural extension of their predecessors in new keynesian framework - Representative Agent New Keynesian models (RANK). Compared to their representative agent counterpart as in e.g. Galí

(2015) or Woodford (2003), these models feature an endogenous heterogeneity of households¹ in form of idiosyncratic shocks to their productivity. Not allowing agents to fully insure against these shocks to their income is a central point of these models and my thesis. Without full insurance they are no longer isomorphic to RANK, which begets implications both economic and numerical. Difference of agents' optimal functions in equilibrium causes distributional aspects to matter for policy analysis and things like Ricardian equivalence no longer hold. This also means that we have new equilibrium object we need to track: measure of agents over states, which is infinitely dimensional and by that as Ben Moll notices is an extreme case of curse of dimensionality and a challenge to rational expectations hypothesis Moll (2026). Solving for equilibrium thus requires new numerical methods not normally taught in core macro courses. Secondary goal of this thesis is exposing the author to these methods.

Despite seminal papers for HANKs being quite recent: Kaplan et al. (2018), Auclert et al. (2024), heterogeneous agents are not new to the profession. They have been known and used in a rigorous way since at least the early 1990s and are often regarded as Bewley-Huggett-Aiyagari models. In his paper, Bewley (1983), Bewley analyses existence of monetary equilibria in an economy where agents hold interest-bearing fiat money and face idiosyncratic shocks. Huggett, in his paper Huggett (1993), considers endowment economy with incomplete markets, where agents can hold risk-free asset. Notably he also allows for borrowing (negative asset holding) up to certain limit and manages to numerically solve and calibrate the model in an effort to explain the equity premium puzzle proposed by Mehra and Prescott (1985). Aiyagari (1994) on the other hand extends this to a production economy where asset is physical capital used by firms in production. Other important early heterogeneous agents paper is Krusell and Anthony A. Smith (1998) where authors allow for aggregate shocks. Solving a model with both aggregate and idiosyncratic shocks is not an easy task, especially so in the 90s. To solve the model without households having to track whole wealth distribution they allow them to make decisions based only on average wealth.

A recent strand of work motivates the questions asked here. Blanchard (2019) in his AEA presidential address argues that when the safe rate is expected to stay below the growth rate - historically the norm rather than the exception in the United States - debt rollovers without any future tax increase may be feasible, so that public debt can carry little or no fiscal cost; he also notes that because the safe rate is the risk-adjusted return to capital, the welfare cost of crowding out may be smaller than usually assumed. Angeletos et al. (2024) ask whether deficits can finance themselves in a new keynesian economy in which Ricardian equivalence fails through finite lives or liquidity constraints; they show that deficits contribute to their own financing through a boom in activity that widens the tax base and a rise in inflation that erodes nominal debt, and that this force strengthens the longer fiscal adjustment is postponed, reaching full self-financing

¹some recent models also heterogenize firm side of the economy

in the limit when monetary policy does not lean too hard against the stimulus. Kaplan et al. (2023) study price-level and inflation dynamics in a heterogeneous-agent economy with nominal government bonds; among their results, the high-debt steady state has lower inflation and Pareto-dominates the low-debt one, a point I return to when discussing equilibrium multiplicity. Finally, Auclert et al. (2025) frame the determination of the long-run real rate as a “race” between asset supply and asset demand: studying the United States, they find that from 1950 to today demand - driven by population aging, rising inequality, foreign demand for U.S. assets, and slower productivity growth - outpaced supply and pushed rates down, but that on current tax and benefit paths rising entitlement costs may let supply win the race going forward. The model in this thesis is a deliberately minimal Bewley-Huggett version of the same asset-market logic, in which the deficit a government can sustain is exactly the supply that clears against precautionary asset demand.

2. Model

Time is discrete, $t \in \{0, 1, \dots, +\infty\}$. The model is an endowment economy à la Huggett consisting of households and the government.

2.1. Households

There is a continuum of infinitely-lived agents indicated by $i \in [0, 1]$. Each of them maximizes their respective expected discounted utility from consumption c_t^i :

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t^i),$$

where $\beta \in (0, 1)$ is a discount factor expressing agent's time preference. We assume functional form of utility function with constant relative risk aversion (CRRA):

$$u(c_t^i) = \frac{(c_t^i)^{1-\gamma} - 1}{1-\gamma},$$

where γ is a CRRA coefficient expressing households preference to smooth out consumption. Every period agent i also receives labor endowment (productivity) y_t^i following AR(1) process:

$$\begin{aligned} \ln(y_t^i) &= \rho_y \ln(y_{t-1}^i) + \epsilon_t, \\ \epsilon_t &\stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_y^2), \end{aligned}$$

where $\rho_y \in (0, 1)$ is persistence of productivity process.

Endowments fluctuate randomly, therefore, every period agents choose whether to save (borrow) by buying (selling) interest-bearing assets maturing next period a_{t+1}^i or consume c_t^i . In doing so they face budget constraint limiting spending to acquired earnings minus lump-sum taxes T_t^i imposed by the government:

$$c_t^i + a_{t+1}^i = (1 + r_t)a_t^i + y_t^i - T_t^i,$$

where y_t^i is endowment of agent i in period t . In period t agent i earns income $(1 + r_t)a_t^i$ from assets a_t^i bought in period $t - 1$ and maturing in period t and receives endowment y_t^i . If agent chooses to borrow he is faced with borrowing constraint $\phi > 0$ limiting degree to which he can be indebted:

$$a_{t+1}^i \geq -\phi.$$

With that we can formulate households optimization problem:

$$\begin{aligned}
& \max_{\{c_t^i, a_{t+1}^i\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{(c_t^i)^{1-\gamma} - 1}{1-\gamma} \\
& \text{s.t.} \\
& c_t^i + a_{t+1}^i = (1 + r_t)a_t^i + y_t^i - T_t^i \quad (\text{budget constraint}) \\
& a_{t+1}^i \geq -\phi \quad (\text{borrowing constraint})
\end{aligned} \tag{1}$$

Taking advantage of problem's recursive nature it can be rewritten as stochastic dynamic programming problem with associated Bellman equation:

$$\begin{aligned}
V(a, y) &= \max_{c_t^i, a_{t+1}^i} \{u(c_t^i) + \beta \mathbb{E} [V_{t+1}(a_{t+1}^i, y_{t+1})|y_t]\} \\
&\text{s.t.} \quad c_t^i + a_{t+1}^i = (1 + r_t)a_t^i + y_t^i - T_t^i \\
&\quad a_{t+1}^i \geq -\phi
\end{aligned}$$

In case of steady state:

$$\begin{aligned}
V(a, y) &= \max_{c, a'} \{u(c) + \beta \mathbb{E} [V(a', y')|y]\} \\
&\text{s.t.} \quad c + a' = (1 + r)a + y - T \\
&\quad a' \geq -\phi
\end{aligned} \tag{2}$$

2.2. Government

We denote taxes collected in period t by T_t , government spending by G_t and one-period bonds issued at $t - 1$, maturing at t , on which the holder receives gross return $(1 + r_t)$ in period t by B_t . In every period government needs to be on its intertemporal government budget constraint (IGBC):

$$B_{t+1} = (1 + r_t)B_t + \underbrace{G_t - T_t}_{-S_t}, \tag{3}$$

where S_t is primary surplus in period t .

2.3. Some government debt arithmetic

Intuition behind (3) is simple: debt repayment is financed by primary surplus and issuance of new debt. Let's assume certain GDP growth $1 + g_{t+1} = \frac{y_{t+1}}{y_t}$. After rewriting it in terms of ratios to GDP (in this case aggregated endowments), in particular $b_t = \frac{B_t}{y_{t-1}}$:

$$b_{t+1} = \frac{1 + r_t}{1 + g_t} b_t - s_t \tag{4}$$

Iterating it forward we get:

$$b_t = \sum_{j=0}^{\infty} \prod_{s=0}^j \left(\frac{1 + g_{t+s}}{1 + r_{t+s}} \right) s_{t+j} + \underbrace{\lim_{T \rightarrow \infty} \prod_{s=0}^{T-1} \left(\frac{1 + g_{t+s}}{1 + r_{t+s}} \right) b_{t+T}}_{=0} \quad (5)$$

We see that in deterministic economy current debt must be equal to the sum of discounted future surpluses plus "bubble" term that we assume goes to zero (No-Ponzi game). Assuming constant surplus and constant positive interest rate r and constant GDP growth g , (5) or (4) give us value of steady-state debt b^* with certain surplus:

$$b^* = \frac{s(1 + g)}{r - g}$$

or alternatively surplus needed to sustain current level of debt:

$$s^* = \frac{r - g}{1 + g} b \quad (6)$$

For $r < g$ it's no longer the case that debt is discounted sum of future surpluses since both terms in (5) diverge. Essentially, when $r < g$ government no longer needs to maintain surpluses since any finite level of debt will eventually repay itself thanks to the growth of economy (we say that economy will grow out of debt). To see it we can look at difference equation we get from (4) setting constant r, g and no surplus $s = 0$:

$$b_{t+1} = \underbrace{\frac{1 + r}{1 + g}}_{<1} b_t$$

Furthermore, we can have permanent deficits ($s < 0$) as long as deficit does not increase debt faster than $\frac{g-r}{1+g}$:

$$-s_t = b_{t+1} - \underbrace{\frac{1 + r}{1 + g}}_{<1} b_t \quad (7)$$

In our model it translates to environment with negative interest rate, since ergodic mean of AR(1) process for logarithm of endowments is zero, aggregated endowments in this economy are always normalized to one, $y_{t+1} = y_t = 1, g_t = g_{t+1} = 0$. How can government perpetuate this Ponzi scheme? Up to this point we abstracted from equilibrium effects - level of debt influences interest rate since asset market needs to clear. Debt issued by government provides supply of assets in asset market, $b_{t+1} = \mathcal{A}_t^s$, where $\mathcal{A}_t^s$ denotes supply of assets. From intertemporal government budget constraint (4) we can derive that asset supply in period t is:

$$\mathcal{A}_t^s(r) = (1 + r)b_t - s_t$$

and in the case of steady-state:

$$\mathcal{A}^s(r) = \frac{s}{r} \quad (8)$$

Given strong precautionary savings motive high asset demand $\mathcal{A}^d$, as we will see later, will be able to sustain $r < g$ environment in steady state for certain levels of fiscal deficit.

2.4. Equilibrium

We define asset space $\mathcal{A} = [-\phi, +\infty)$, endowment space $\mathcal{Y} = \mathbb{R}$ and state space $\mathcal{A} \times \mathcal{Y}$. **Stationary recursive competitive equilibrium** is a value function $V(a, y)$, policy functions $a'(a, y)$ and $c(a, y)$, interest rate r and distribution $\mu(a, y)$ such that given government surplus s , spending G and tax system $T(a, y)$:

- given r , value function $V(a, y)$ satisfies Bellman equation (2) and associated policy functions $a'(a, y)$ and $c(a, y)$ solve households optimization problem (1),
- $\mu(a, y)$ is stationary distribution, i.e. for every borel set $B \in \mathcal{B}(\mathcal{A} \times \mathcal{Y})$:

$$\mu(B) = \int_{\mathcal{A} \times \mathcal{Y}} P((a, y), B) d\mu(a, y),$$

where $P(\cdot)$ is transition function induced by households AR(1) process and policy function $a'(a, y)$,

- markets clear:

$$\underbrace{\int_{\mathcal{A} \times \mathcal{Y}} a'(a, y) d\mu(a, y)}_{\mathcal{A}^d} = \underbrace{\frac{s}{r}}_{\mathcal{A}^s} \quad (\text{asset market})$$

$$\int_{\mathcal{A} \times \mathcal{Y}} y d\mu(a, y) = \int_{\mathcal{A} \times \mathcal{Y}} c(a, y) d\mu(a, y) + G \quad (\text{goods market})$$

- government budget constraint is satisfied:

$$s = \int_{\mathcal{A} \times \mathcal{Y}} T(a, y) d\mu(a, y) - G$$

3. Calibration and numerical solution

Our model is solved by discretizing the state space. Due to assumed high persistence of AR(1) process for idiosyncratic productivity we approximate a 5 state Markov chain using Rouwenhorst method (Rouwenhorst (1995)). State values are then normalized so that mean productivity is unity. Asset space is discretized into a non-uniform grid with 500 points. Points are chosen using polynomial expansion with parameter $\theta = 5$ so that most grid points concentrate near borrowing constraint. Bellman equation associated with household optimization problem is solved using Endogenous Grid Method (Carroll (2006)). Lastly, to calculate stationary distribution in equilibrium we use method by Young (Young (2010)), which ensures no bias of aggregate values.

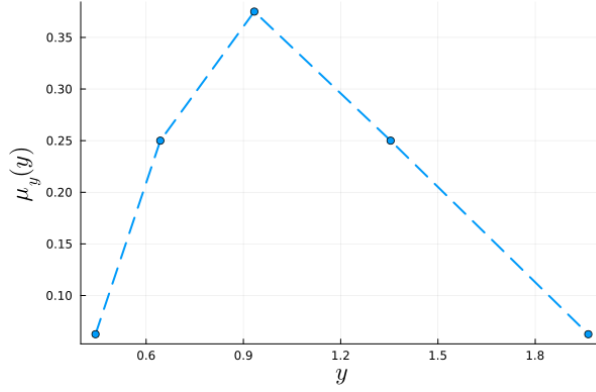


Figure 2: Income levels discretized with $N_z = 5$ states using Rouwenhorst method.

Model parameters	Description	Value
β	Discount factor	0.96
γ	CRRA coefficient	2.0
ρ_y	income persistence	0.897
σ_y	Income innovation volatility	$\sqrt{0.027}$
ϕ	Borrowing constraint	0.175
Numerical parameters	Description	Value
N_z	Number of income states	5
a_{max}	Maximum assets	30.0
N_a	Asset grid points	500
θ	Grid curvature parameter	5.0

Table 1: Baseline model calibration and numerical parameters.

Assumed annual discount factor is 0.96 and CRRA coefficient is 2.0 - standard values for models of countries like Poland. Income persistence and volatility is taken from Kolasa (2017) and converted to annual values. Chosen borrowing constraint for Poland is 0.175, which is lower than 0.25 for US in Kaplan et al. (2018). Unfortunately there is no reliable and actual data on borrowing constraint so calibration should be viewed only as an illustration.

4. Main results

4.1. Maximum permanent deficit

We begin with overview of driving forces in asset market and possible steady state equilibria. If $r = \beta^{-1} - 1$ without idiosyncratic shocks households would be indifferent between consumption today and tomorrow. With this interest rate value, in presence of uninsurable shocks,

precautionary savings motive would drive asset demand to infinity. This means that the upper bound on interest rate is $r = \beta^{-1} - 1$ and there is horizontal asymptote there. Furthermore, as interest rate decreases it is less costly to borrow and lending becomes less profitable, i.e. more and more households become borrowers instead of lenders. This means that asset demand is increasing in interest rate and there is a lower bound on assets at $\mathcal{A} = -\phi$, where all households are at their borrowing limit.

If on the asset market there are only households, asset supply is zero since amount of borrowing and lending between households nets out in aggregation. Given sufficiently strong precautionary savings motive interest rate at zero asset supply, denoted $\underline{r}$, can be negative. For government it means possible fiscal space for permanent deficits financed by households wanting to insure themselves against idiosyncratic shocks, as seen on figure 3.

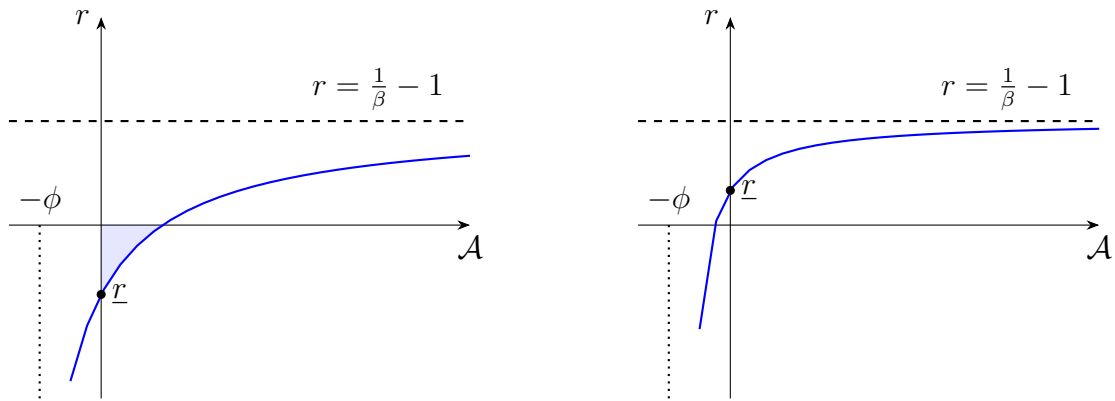


Figure 3: Asset demand curve that permits (left) and does not permit (right) deficits in stationary equilibrium.

Relation between deficits $-s$, government debt b and interest rate r is given by $b = \frac{s}{r}$. As seen on figure 4 for sufficiently small deficits there is a problem of equilibrium multiplicity. There are two steady states, first one with more negative interest rate and therefore lower steady-state debt and second with higher interest rate and also higher steady-state government debt. Kaplan et al. (2023) consider a richer, nominal setting with flexible prices, where government bonds are nominal. They prove that higher debt steady state equilibrium has lower inflation and Pareto dominates low debt equilibrium. Additionally, high debt steady state is saddle-path stable, meanwhile many initial equilibria lead to lower debt steady-state. This poses a problem for price level determination, which can be resolved by eliminating low debt steady-state. Price level determination can be restored in several ways. Government can follow a debt reaction rule, increasing deficits if debt is higher than in saddle path steady state. It can also have real rate reaction rule, where it increases deficits if real rate is too low. Although these rules may seem counterintuitive they provide uniqueness. Alternatively uniqueness may also be provided by introducing foreign demand for assets that is sufficiently inelastic.

From figure 4 we also see an interesting "knife-edge" case where asset demand and supply

curves are tangent to each other and deficit achieves its maximum value. If it were any higher, debt would have explosive dynamics, since debt increase from deficits would overcome decrease coming from debt repayment below it's borrowed value.

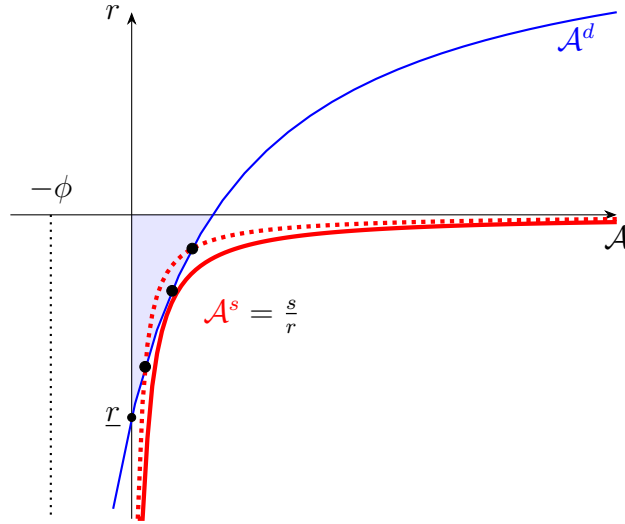


Figure 4: Equilibria on asset market with different levels of deficit.

4.2. Maximum deficit with no taxes

We start our analysis with a simple model illustrating possibility of permanent budget deficit. Since taxes (transfers) affect households optimization problem and in turn aggregated asset demand curve we decide to abstract from them, at least for now, to focus on: the deficit, the role of government as bond issuer, households policy functions and nature of asset demand curve in this class of models. Here government is only a bond issuer whose deficits finance certain unproductive government spending G . To illustrate multiplicity of equilibria we initially assume government spending amounting to $G = 0.4\%$ of GDP i.e. primary surplus $s = -0.4\%$. Everything else calibrated as described in section two. From figure 5 we see that if there were no government, asset supply would be zero, resulting in steady state interest rate $\bar{r} = -4.35\%$. With this level of deficit we also managed to find two equilibria: $r_1^* = -2.54\%$ corresponding to debt at $B_1^* = 15.75\%$ of GDP and $r_2^* = -0.92\%$ with $B_2^* = 43.48\%$ of GDP. Before overiewing other equilibrium objects we run a bisection on deficit levels between 0.4% and 0.5% , looking for deficit level for which two equilibria would be sufficiently close and result in maximum deficit.

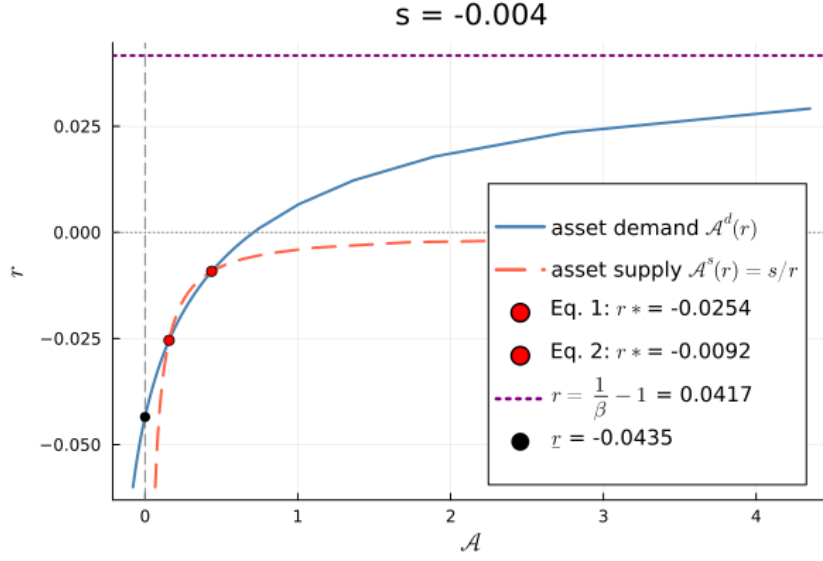


Figure 5: Two stationary equilibria with $s = -0.4\%$.

We manage to find such a deficit level with very high precision, although interest rates for two steady states are still identical only to third decimal place. Further increasing precision required from bisection mostly increases algorithms runtime and as it turns out mean of both interest rates is a fairly good approximation of interest rate for unique equilibrium with maximum deficit. As can be seen on figure 6 value of deficit is 0.4738% for interest rate $r^* = -1.65\%$ and corresponds to debt level $B^* = 28.73\%$.

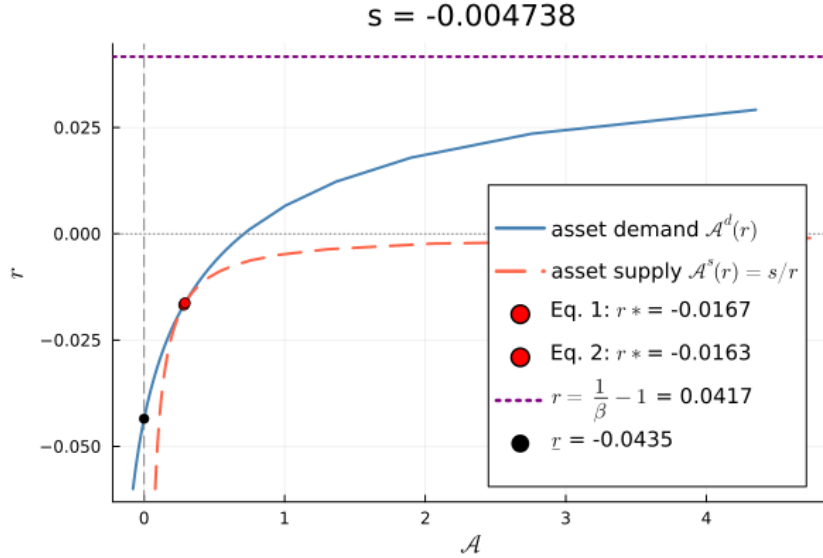


Figure 6: Maximum deficit in economy without taxation.

Now a little bit more about Huggett model itself. Below, on figure 7 we can see policy functions of households for different levels of endowments for steady state corresponding to maximum deficit with $r^* = -1.65\%$. There is also a 45° line for easier interpretation, if pol-

icy function is below this line, households decrease their asset holdings, if above, their asset holdings increase. We see that for households with highest endowment their optimal policy is to save and build up wealth up to around five times mean endowment. On the other hand for households with lowest endowment there is, although hard to be seen from this figure, an area around the borrowing constraint where their optimal policy is to borrow up to that limit. Those are exactly the so called Hand-to-Mouth households that are not on their respective Euler equation and have the highest values of marginal propensity to consume, which plays a crucial role in monetary policy transmission. For lowest and most of other endowment levels it is generally optimal to consume part of their asset wealth.

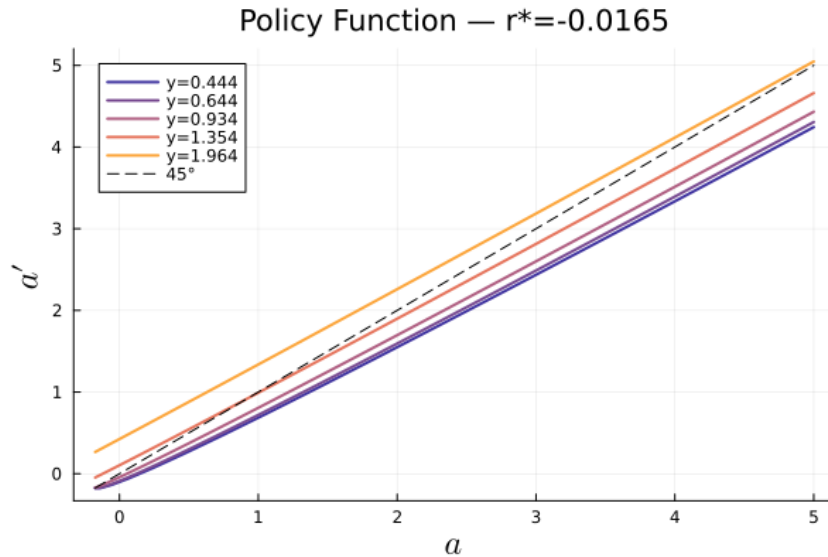


Figure 7: Households policy functions with unique equilibrium at $r^* = -0.0165$.

With optimal policy functions we are also able to find stationary distribution of assets in our steady state, seen on figure 8. It can immediately be observed that most of households are at borrowing constraint and that distribution is right-skewed, i.e. there is a long right tail of households with substantial levels of wealth. There can also be seen spikes caused by the fact that endowment levels were discrete.

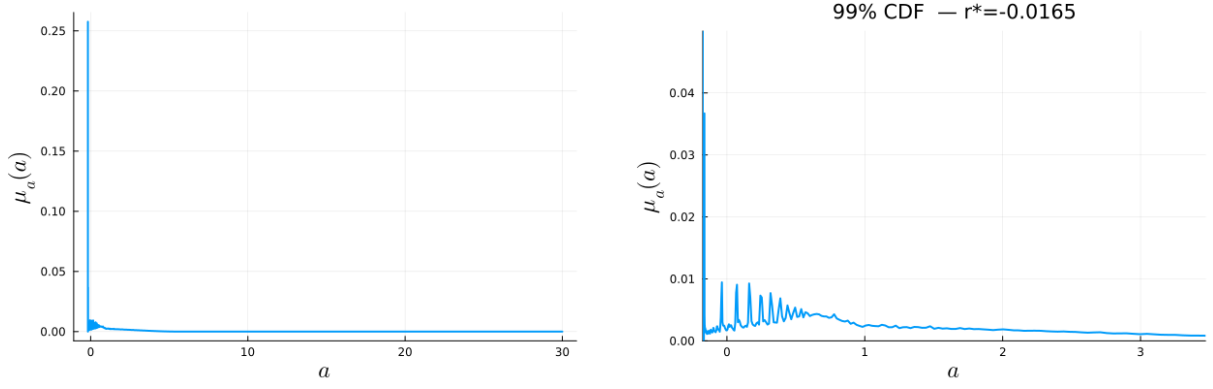


Figure 8: Asset distribution in stationary equilibrium with $r^* = -0.0165$.

4.3. Welfare analysis of equilibria

Last thing we do with these results is compare welfare in equilibria we found. The intuition is that a higher amount of assets in equilibrium will result in higher welfare, thanks to ability of households to better insure themselves against income fluctuations. Welfare is defined as integral of households value function over possible states:

$$\mathcal{W} = \int_{\mathcal{A} \times \mathcal{Y}} V(a, y) d\mu(a, y).$$

We used value function $V(a, y)$ because its economic interpretation is maximum expected life-time discounted utility one can achieve with his current state being (a, y) . Since welfare itself may seem abstract we rewrite it in terms of consumption equivalent variation (CEV):

$$\lambda_k^j = \left(\frac{\mathcal{W}^j}{\mathcal{W}^k} \right)^{\frac{1}{1-\gamma}} - 1,$$

which itself is interpreted as a percent increase of everybody's consumption under policy k required to achieve same level welfare as under policy j . We start by comparing low debt equilibrium (equilibrium 1) and high debt equilibrium (equilibrium 2) in case of 0.4% deficit. From figure 9 it can be noticed that with higher level of debt households are able to accumulate more wealth, corresponding to "fatter" right tail in asset distribution. The reason for that can be observed from comparison of policy function. For higher debt level households with high endowment decide to accumulate bigger buffer stock of assets or alternatively consume less out of it when endowment is low than they would have did when debt level is low. Consumption equivalent variation is 6.8%, meaning that ceteris paribus, for welfare to be the same in both steady states, every endowment in low debt steady state should be 6.8% higher. This analysis has confirmed that although aggregate income in this economy is fixed having more assets can be welfare improving through better smoothing out effects of idiosyncratic income fluctuations. For that reason some researchers approximate heterogeneous models with adding bonds into utility function of representative agent.

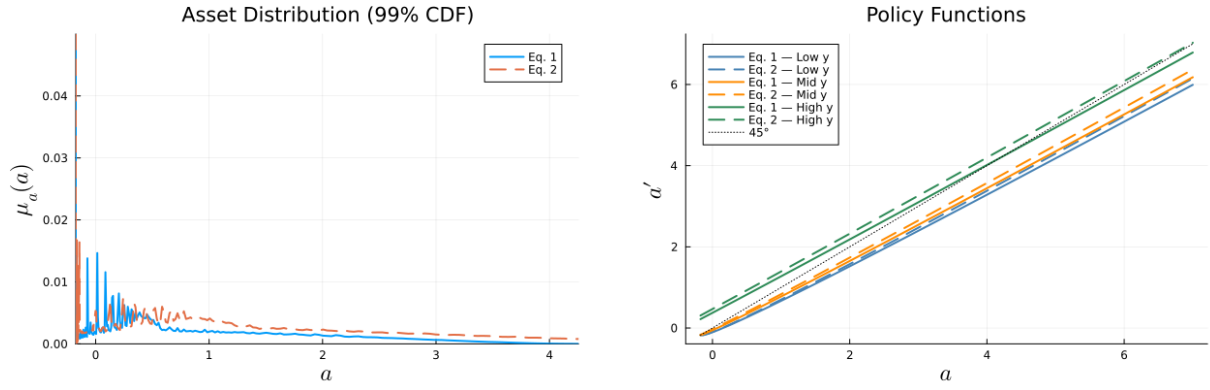


Figure 9: Comparison of asset distributions and policy functions of low debt equilibrium - "eq. 1" and high debt equilibrium - "eq. 2".

Next we do even more insightful comparison. We compare equilibrium with no government to that where its sole role is issuance of debt, for which it is able to finance its spending. We assume maximum deficit. Before turning to the numerical results, it is worth reflecting on the mechanism allowing this works. If existence of an agent whose only role is to issue debt and even more, who has a certain rent (deficit) from it, is welfare improving, then why haven't households done it on their own? Situation is similar to that of OLG economy considered by Diamond (1965) and later Blanchard (2019). In their analysis OLG economy is dynamically inefficient since agents overaccumulate physical capital to save for their retirement. Introduction of pay-as-you-go transfers from government or equivalently government debt rollover would then be welfare improving. Similarly in Huggett model households wanting to buy assets are limited by asset supply and borrowing limit of other side of transaction. Government debt fixes this and is *de facto* a transfer b_{t+1} from current period t households wanting to ensure themselves against future income fluctuations to households that did the same in previous period $t - 1$. For facilitating this government receives in period t rent equal to $-r_t b_t$ and by that households receive only $(1 + r_t)b_t < b_{t+1}$.

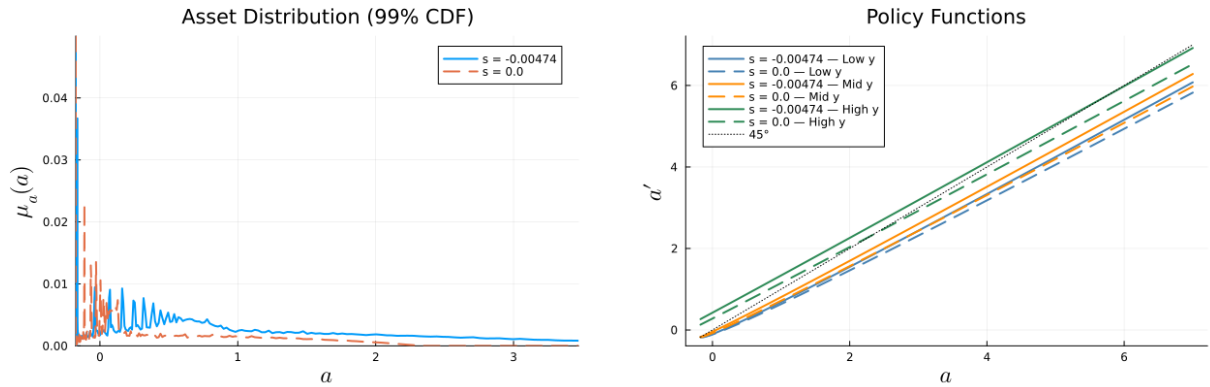


Figure 10: Comparison of asset distributions and policy functions of model with no government to maximum deficit case.

From figure 10 it can be noticed that government debt allows households to accumulate more assets as buffer, as for welfare implication, consumption equivalent variation in this case is $\lambda = 4.1\%$. It means that thanks to government debt welfare is as if they received endowments 4.1% higher.

4.4. Taxes and asset demand

In this section we investigate how tax system affects precautionary motive and by that fiscal space of the government. In principle, as Kaplan et al. (2023) notice, more redistributive tax and transfer system decreases fiscal space. For example, lump-sum taxes compared to progressive or proportional taxation of the same aggregate value are more of a burden for low income taxpayers. By amplifying or reducing precautionary savings motive government affects aggregate demand curve. For simplicity we choose a case of lump-sum tax levied on all households. Assuming initially lump sum taxes $t = 30\%$ and primary surplus $s = -2\%$, asset demand and supply curves can be seen on figure 11. We see that compared to the case of no taxation, interest rate $\bar{r}$ with no asset supply (balanced government budget) decreased from more than -5% to less than -8% and by that fiscal space has increased.

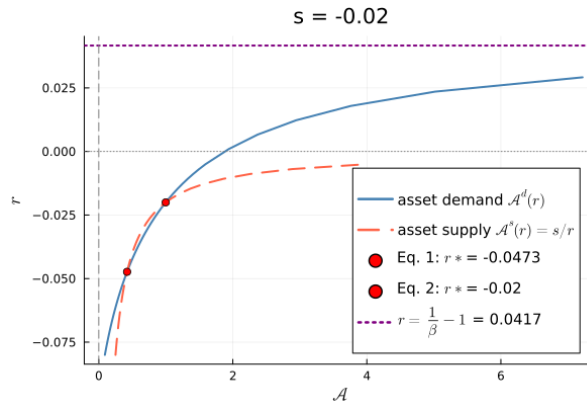


Figure 11: Two stationary equilibria with $t = 30\%$, $s = -2\%$.

Maximum deficit with this tax system is 2.22% and by that is more than four times higher than with no taxes at all. Although this example is fairly simple, it implies that assumed tax system and its redistributive power is crucial for fiscal space of government.

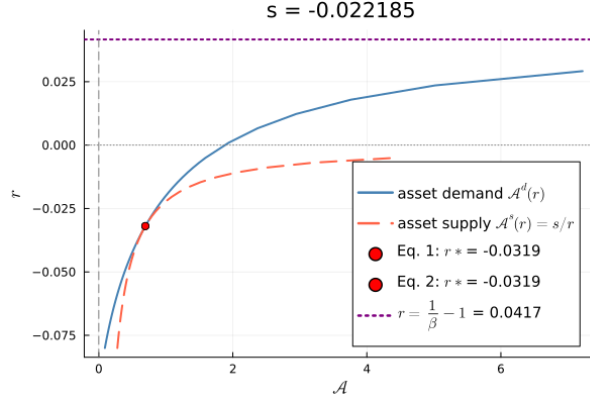


Figure 12: Maximum deficit in economy with lump-sum taxes $t = 30\%$.

5. Conclusions

This thesis has used a Bewley-Huggett economy to analyze how large a permanent primary deficit can a government run when households face uninsurable idiosyncratic income risk and a binding borrowing constraint. Because income shocks cannot be fully insured, households build up a buffer stock of savings for precautionary reasons; this generates a demand for a safe store of value that the private sector, where every loan is someone else's debt, cannot satisfy in the aggregate. Government bonds fill that gap. The steady-state real interest rate is then whatever level clears the asset market, and when the precautionary motive is strong enough that rate is driven below the economy's growth rate. In this region the arithmetic of the intertemporal budget constraint changes character: a finite stock of debt no longer has to be repaid out of future surpluses, and the government can sustain a strictly positive deficit forever, financed not by promise of future austerity but by households' permanent willingness to hold its debt as insurance. The deficit is, in effect, the flow return the government earns for supplying liquidity that households value and cannot manufacture themselves.

For small deficits there are two stationary equilibria, a low-debt one with a more negative interest rate and a high-debt one with a less negative rate. As the deficit rises these two equilibria move toward each other, until at a knife-edge level the asset-supply and asset-demand curves become tangent and the two roots merge into one. Any deficit larger than this has no stationary equilibrium at all, the increase in debt coming from the primary deficit overwhelms the erosion of debt coming from the sub-growth interest rate and debt dynamics become explosive. In the baseline calibration without taxes this maximum permanent deficit is about 0.47% of GDP, supported at a real rate of roughly -1.65% and a debt level near 29% of GDP.

Second, government debt is welfare-improving in this environment, and measurably so. Comparing the low- and high-debt equilibria at a common deficit yields a consumption-equivalent gain of about 6.8%, while comparing an economy with no government to the maximum-deficit

economy yields about 4.1%. The source of the gain is not higher aggregate income, output is fixed by endowments, but better smoothing of consumption by households that face fluctuations to their income. A larger stock of public debt lets households hold bigger buffers and rely less on the borrowing constraint. This is the incomplete-markets counterpart of the dynamic-inefficiency logic familiar from Diamond (1965) and Blanchard (2019). Just as pay-as-you-go transfers can raise welfare in an over-accumulating overlapping-generations economy, government debt raises welfare here by providing liquidity that households cannot create when left on their own.

Third, the tax system is a primary determinant of fiscal space. Introducing a lump-sum tax, which falls proportionately harder on low-income households than a progressive or proportional levy of the same aggregate value, sharpens the precautionary motive precisely among the households closest to the borrowing constraint. This shifts the asset-demand curve and raises the maximum sustainable deficit to about 2.2% of GDP, more than four times the no-tax value.

References

- Aiyagari, S. R. (1994). Uninsured idiosyncratic risk and aggregate saving. *The Quarterly Journal of Economics*, 109(3):659–684.
- Angeletos, G.-M., Lian, C., and Wolf, C. K. (2024). Can deficits finance themselves? *Econometrica*, 92(5):1351–1390.
- Auclert, A., Malmberg, H., Rognlie, M., and Straub, L. (2025). The race between asset supply and asset demand. Working Paper 34470, National Bureau of Economic Research.
- Auclert, A., Rognlie, M., and Straub, L. (2024). The intertemporal keynesian cross. *Journal of Political Economy*, 132(12):4068–4121.
- Bewley, T. (1983). A difficulty with the optimum quantity of money. *Econometrica*, 51(5):1485–1504.
- Blanchard, O. (2019). Public debt and low interest rates. *American Economic Review*, 109(4):1197–1229.
- Carroll, C. D. (2006). The method of endogenous gridpoints for solving dynamic stochastic optimization problems. *Economics Letters*, 91(3):312–320.
- Diamond, P. A. (1965). National debt in a neoclassical growth model. *The American Economic Review*, 55(5):1126–1150.
- Galí, J. (2015). *Monetary Policy, Inflation, and the Business Cycle: An Introduction to the New Keynesian Framework and Its Applications*. Number 10495 in Economics Books. Princeton University Press, 2 edition.
- Huggett, M. (1993). The risk-free rate in heterogeneous-agent incomplete-insurance economies. *Journal of Economic Dynamics and Control*, 17(5):953–969.
- Kaplan, G., Moll, B., and Violante, G. L. (2018). Monetary policy according to hank. *American Economic Review*, 108(3):697–743.
- Kaplan, G., Nikolakoudis, G., and Violante, G. L. (2023). Price level and inflation dynamics in heterogeneous agent economies. Working Paper 31433, National Bureau of Economic Research.
- Kolasa, A. (2017). Life cycle income and consumption patterns in poland. *Central European Journal of Economic Modelling and Econometrics*, 9(2):137–172.

- Krusell, P. and Anthony A. Smith, J. (1998). Income and wealth heterogeneity in the macroeconomy. *Journal of Political Economy*, 106(5):867–896.
- Mehra, R. and Prescott, E. C. (1985). The equity premium: A puzzle. *Journal of Monetary Economics*, 15(2):145–161.
- Moll, B. (2026). The trouble with rational expectations in heterogeneous agent models: A challenge for macroeconomics. *The Economic Journal*, 136(676):1173–1216.
- Rouwenhorst, K. G. (1995). *Asset Pricing Implications of Equilibrium Business Cycle Models*, pages 294–330. Princeton University Press.
- Woodford, M. (2003). *Interest and Prices: Foundations of a Theory of Monetary Policy*. Princeton University Press.
- Young, E. R. (2010). Solving the incomplete markets model with aggregate uncertainty using the krusell-smith algorithm and non-stochastic simulations. *Journal of Economic Dynamics and Control*, 34(1):36–41.

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A. Numerical errors

This section provides Euler residuals of models presented in the main body of text.

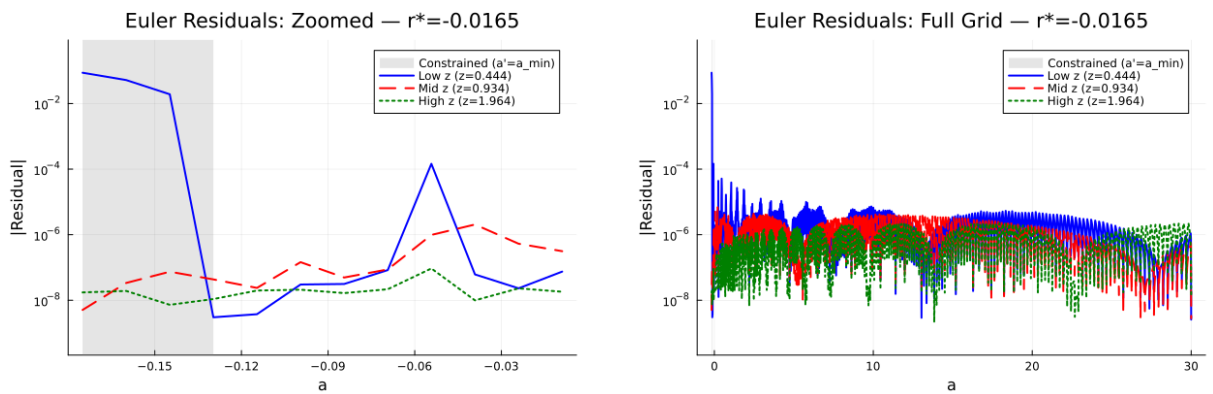


Figure 13: Euler residuals of model with no taxes.

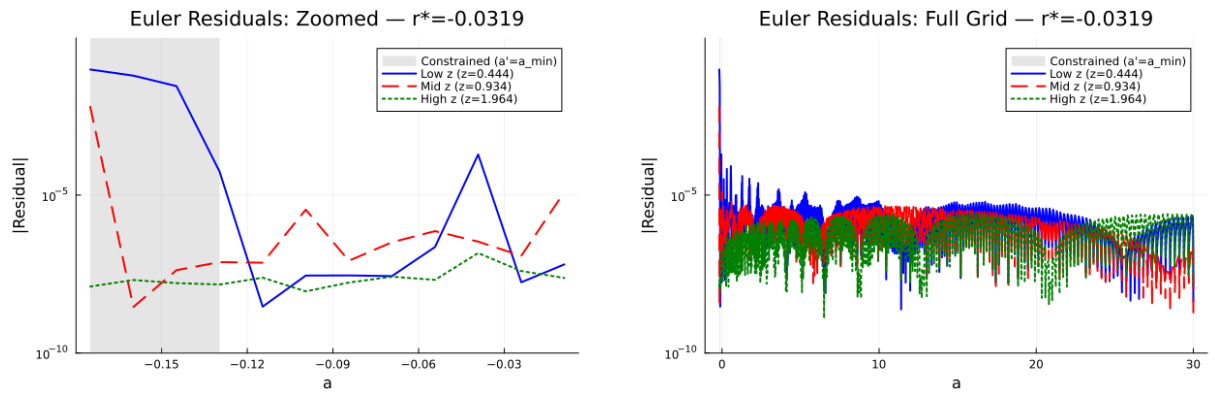


Figure 14: Euler residuals of model with taxes.

Summary

This thesis studies how large a permanent budget deficit a government can sustain in a Bewley–Huggett economy where households face uninsurable income shocks and a borrowing constraint. Precautionary saving generates demand for a safe asset, which government bonds supply. The steady-state interest rate clears the asset market and can fall below the growth rate, allowing a permanent deficit financed by households’ willingness to hold debt rather than by future surpluses. The maximum sustainable deficit, at which asset supply and demand are tangent, in the no-tax calibration is about 0.47% of GDP. Government debt is welfare-improving through better consumption smoothing, with consumption-equivalent gains of 4 – 7%. A lump-sum tax, which strengthens the precautionary motive, raises the maximum deficit fourfold to about 2.2% of GDP, showing that fiscal space depends on how redistributive the tax system is.